LOCAL-PROMPT: EXTENSIBLE LOCAL PROMPTS FOR FEW-SHOT OUT-OF-DISTRIBUTION DETECTION

Anonymous Author

ABSTRACT

Out-of-distribution (OOD) detection is critical for safely deploying visual recognition systems, yet acquiring large labeled datasets for this purpose is often impractical. Vision-language models such as CLIP offer a promising foundation for few-shot OOD detection through prompt learning, but existing approaches primarily optimize global prompts that capture holistic image semantics while overlooking fine-grained regional cues essential for distinguishing subtle, near-distribution outliers. We propose Local-Prompt, an extensible framework that freezes pretrained global prompts and exclusively trains local (regional) prompts through a coarse-to-fine scheme. Our method introduces two key components: (1) global prompt guided hard-negative augmentation, which leverages frozen global prompts to select challenging positive and negative crops without requiring real OOD data, and (2) local prompt enhanced regional regularization, which trains class-specific local prompts alongside a bank of local negative prompts via contrastive and diversity objectives over top-k image regions. At inference, a Regional-MCM score seamlessly combines global and local evidence, and the framework naturally accommodates plug-in integration with independently trained global prompts. On the ImageNet-1k benchmark with only 4-shot supervision, Local-Prompt combined with LoCoOp achieves an average FPR95 of 26.29% and AUROC of 94.03%, surpassing prior methods that use substantially more labeled data.

AI-Generated

1 INTRODUCTION

Deploying machine learning models in open-world environments requires the ability to detect out-of-distribution (OOD) inputs—samples that do not belong to any class seen during training Hendrycks & Gimpel (2016). While OOD detection has been extensively studied in the fully supervised setting Liu et al. (2020); Sun et al. (2022); Hendrycks et al. (2018), the cost of collecting large-scale labeled datasets motivates few-shot approaches that achieve robust detection with minimal supervision.

Vision-language models (VLMs) such as CLIP Radford et al. (2021) have emerged as a powerful foundation for OOD detection due to their rich, transferable representations learned from large-scale image-text pairs. Zero-shot methods like MCM Ming et al. (2022) and GL-MCM Miyai et al. (2023b) exploit CLIP's alignment between visual and textual features to perform OOD detection without any task-specific training. Few-shot prompt learning methods such as CoOp Zhou et al. (2022b) and LoCoOp Miyai et al. (2023a) further improve performance by optimizing continuous prompt vectors with a small number of labeled examples. However, these approaches primarily operate on global image features and optimize prompts at the holistic level, which limits their capacity to capture fine-grained, region-level signals that are often the distinguishing factor for near-OOD samples.

Consider the task of distinguishing a golden retriever (in-distribution) from a Labrador retriever (OOD): the discriminative cues are often confined to subtle local regions such as coat texture or ear shape, rather than the global image composition. Existing global prompt methods tend to produce similar representations for such pairs, leading to elevated false positive rates on hard OOD cases. While GL-MCM Miyai et al. (2023b) introduced the use of local image features at inference time, it does not learn local prompts, leaving local feature matching under-optimized.

To address this gap, we propose Local-Prompt, a coarse-to-fine framework that freezes pretrained global prompts and exclusively trains local prompts to capture regional discriminative cues. Our approach consists of two complementary components. First, *global prompt guided hard-negative augmentation* generates challenging training crops by leveraging frozen global prompts to identify hard positives and negatives from random augmentations, injecting local outlier knowledge without requiring real OOD data. Second, *local prompt enhanced regional regularization* trains class-specific local prompts and a bank of local negative prompts through contrastive objectives over top-k image regions, encouraging the model to develop sharp region-level discriminability. At inference, we introduce Regional-MCM, a scoring function that aggregates global MCM with top-k regional similarity scores, including contributions from negative prompts. Crucially, Local-Prompt is *extensible*: because global prompts remain frozen during training, independently trained global prompts (*e.g.*, from LoCoOp) can be seamlessly integrated at inference time to further boost performance.

Our main contributions are summarized as follows:

- We introduce a coarse-to-fine few-shot OOD detection paradigm that freezes global prompts and trains local prompts to capture regional outlier cues, enabling extensible plug-in integration with any global prompt method.
- We propose global prompt guided hard-negative augmentation and local prompt enhanced regional regularization with contrastive top-k region matching and negative prompt diversity, providing fine-grained OOD supervision without real outlier data.
- We develop Regional-MCM, a scoring function that combines global and local evidence from both positive and negative prompts for robust OOD detection and ID classification.
- We demonstrate state-of-the-art few-shot OOD detection results on ImageNet-1k and ImageNet-100 benchmarks. With only 4-shot supervision, Local-Prompt combined with LoCoOp achieves an average FPR95 of 26.29% and AUROC of 94.03%, outperforming prior methods including those using 16-shot supervision.

2 Related Work

2.1 Vision-Language Models for OOD Detection

Vision-language models (VLMs) pretrained on large-scale image-text pairs Radford et al. (2021) have shown remarkable zero-shot transfer capabilities, motivating their application to OOD detection. MCM Ming et al. (2022) first demonstrated that the softmax of CLIP's text-image cosine similarities serves as an effective zero-shot OOD score without any training. GL-MCM Miyai et al. (2023b) extended this idea by incorporating both global and local (patch-level) image features, showing that local features provide complementary evidence for detecting subtle OOD samples. CLIPN Wang et al. (2023) proposed learning "no" prompts to explicitly model negative semantics. ZOC Esmaeilpour et al. (2022) leveraged CLIP for zero-shot OOD detection by generating candidate OOD class names. NegLabel Jiang et al. (2024) exploited large external text corpora to mine negative labels that serve as proxies for OOD classes. While these methods effectively leverage CLIP's pretrained knowledge, they either operate in a zero-shot setting without task adaptation or rely on external data sources. In contrast, Local-Prompt adapts CLIP to the target task through few-shot local prompt learning while remaining self-contained.

2.2 Prompt Learning for Vision-Language Models

Prompt learning offers a parameter-efficient approach to adapting VLMs to downstream tasks. CoOp Zhou et al. (2022b) introduced learnable continuous context vectors that replace hand-crafted text prompts, achieving strong few-shot image classification. CoCoOp Zhou et al. (2022a) extended CoOp with input-conditional prompts for improved generalization. Beyond classification, prompt learning has been applied to various vision tasks Bahng et al. (2022); Jia et al. (2022). For OOD detection, LoCoOp Miyai et al. (2023a) proposed regularizing prompt learning by encouraging local image features of ID data to be dissimilar from the learned prompts, thereby pushing OOD-like local regions away from ID class representations. However, LoCoOp optimizes a single set of global prompts and uses local features only as a regularization signal during training, without learning dedicated local prompts. Our method complements these approaches by introducing trainable local

AI-Generated

prompts that directly optimize region-level matching, and our frozen-global design means Local-Prompt can be combined with any global prompt method at inference.

2.3 Few-Shot OOD Detection

Few-shot OOD detection aims to build reliable detectors with minimal labeled in-distribution data. Traditional approaches adapted classical OOD scores Hendrycks & Gimpel (2016); Liu et al. (2020); Sun et al. (2022) to the few-shot regime, but performance degraded substantially under limited supervision. Fort *et al*. Fort et al. (2021) studied few-shot OOD detection with pretrained transformers, finding that fine-tuning on even a handful of examples can improve detection. NPOS Tao et al. (2022) proposed non-parametric outlier synthesis in the feature space to provide pseudo-OOD supervision. Outlier Exposure Hendrycks et al. (2018) leveraged auxiliary OOD datasets to regularize model training, though this requires access to external data. ID-like Bai et al. (2024) addressed few-shot OOD detection by mining ID-like hard negatives from CLIP's feature space to sharpen decision boundaries. Our approach shares the goal of generating challenging negatives, but differs in that we synthesize hard negatives through augmentation-based crops guided by frozen global prompts, avoiding reliance on external corpora or feature-space sampling.

2.4 Local Feature Based OOD Detection

Leveraging local or region-level features for OOD detection has gained attention as a means to capture fine-grained anomalies that global representations miss. Park and Chung Park et al. (2023) analyzed the utility of local features from vision transformers for near-OOD detection. GL-MCM Miyai et al. (2023b) combined global and local CLIP features at inference time to improve zero-shot OOD scoring. Zhang *et al*. Zhang et al. (2023) studied the interplay between global and local representations for robust OOD detection. CLIPScope Shu et al. (2023) investigated CLIP-based features across distribution shifts including OOD scenarios. These methods primarily exploit local features at inference without learning task-specific local representations. Local-Prompt advances this line of work by introducing learnable local prompts that are explicitly optimized for regional OOD discrimination, combined with a bank of negative prompts and a diversity loss that encourages coverage of diverse OOD patterns across local regions.

3 Preliminary

CLIP. Contrastive Language-Image Pre-training (CLIP) Radford et al. (2021) learns aligned visual and textual representations through contrastive learning on large-scale image-text pairs. CLIP consists of a visual encoder f_v and a text encoder f_t that map images and text into a shared embedding space. Given an image $\mathbf{x}$, the visual encoder produces a global feature $\mathbf{f}_v(\mathbf{x}) \in \mathbb{R}^d$ as well as a set of local (patch-level) features $\{\mathbf{f}_v^l(\mathbf{x})_i\}_{i=1}^N$, where N is the number of spatial patches. For a text prompt $\mathbf{p}$, the text encoder yields a text feature $\mathbf{f}_t(\mathbf{p}) \in \mathbb{R}^d$. The similarity between an image and text is measured by cosine similarity: $\mathrm{sim}(\mathbf{x}, \mathbf{p}) = \langle \mathbf{f}_v(\mathbf{x}), \mathbf{f}_t(\mathbf{p}) \rangle / (\|\mathbf{f}_v(\mathbf{x})\| \cdot \|\mathbf{f}_t(\mathbf{p})\|)$.

Maximum Concept Matching (MCM). MCM Ming et al. (2022) leverages the zero-shot classification capability of CLIP for OOD detection without any training. Given a set of C in-distribution class names, MCM constructs text prompts $\{\mathbf{p}_c\}_{c=1}^C$ using the template "a photo of a [class]" and computes the OOD score as the negative maximum softmax probability:

$$S_{\mathrm{MCM}}(\mathbf{x}) = -\max_c \frac{\exp(\tau \cdot \langle \mathbf{f}_v(\mathbf{x}), \mathbf{f}_t(\mathbf{p}_c) \rangle)}{\sum_{j=1}^C \exp(\tau \cdot \langle \mathbf{f}_v(\mathbf{x}), \mathbf{f}_t(\mathbf{p}_j) \rangle)}, \tag{1}$$

where τ is the logit scale parameter of CLIP. A lower (more negative) S_{MCM} indicates higher confidence that $\mathbf{x}$ is in-distribution.

Few-Shot OOD Detection. In the few-shot OOD detection setting, we are given a small labeled training set $\mathcal{D}_{\mathrm{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{C \times K}$ containing K labeled samples per class for C in-distribution classes. The goal is to learn a scoring function $S(\mathbf{x})$ that assigns lower scores to in-distribution (ID) test samples drawn from the same distribution as $\mathcal{D}_{\mathrm{train}}$, and higher scores to out-of-distribution (OOD) test samples, without access to any OOD data during training.

AI-Generated

Prompt Learning for OOD Detection. CoOp Zhou et al. (2022b) replaces the hand-crafted text template with learnable context vectors $[\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_M, \mathbf{c}_j]$, where $\{\mathbf{v}_m\}_{m=1}^{M}$ are shared learnable tokens and $\mathbf{c}_j$ is the word embedding of the j-th class name, trained via cross-entropy loss on few-shot labeled data. LoCoOp Miyai et al. (2023a) extends CoOp by introducing an OOD regularization loss computed on local (patch-level) features, encouraging local features of training images to be dissimilar from an entropy-maximized distribution over classes. While LoCoOp improves OOD detection, it relies on a single set of global prompts and does not explicitly model the distribution of OOD-like local regions.

4 METHOD

4.1 OVERVIEW

We propose Local-Prompt, a few-shot OOD detection framework that leverages class-specific local prompts and learnable negative prompts to capture fine-grained regional information for distinguishing ID and OOD samples. Our approach addresses a key limitation of prior prompt-learning methods: they operate primarily at the global image level and fail to exploit the rich local structure available in CLIP's patch-level features.

Local-Prompt operates in two stages. In the first stage (Section 4.2), we perform *Global Prompt Guided Negative Augmentation*, which generates hard positive and hard negative local image features from random crops of training images using frozen global prompts, without requiring any real OOD data. In the second stage (Section 4.3), we train *Local Prompt Enhanced Regional Regularization*, where class-specific local prompts and learnable negative prompts are jointly optimized using the augmented features. At inference time (Section 4.4), we introduce a *Regional-MCM* score that combines global and local prompt-based scores for robust OOD detection.

Our framework introduces three types of text prompts:

- **Global prompts**: Frozen prompts initialized as "a photo of a [class]" embeddings, used to guide the negative augmentation stage.
- **Local prompts**: Class-specific learnable context vectors $[\mathbf{v}_1^l, \mathbf{v}_2^l, \ldots, \mathbf{v}_M^l, \mathbf{c}_j]$ that capture fine-grained class characteristics at the patch level.
- **Negative prompts**: A set of N_{neg} learnable context vectors $\{\mathbf{n}_q\}_{q=1}^{N_{\text{neg}}}$ without class name tokens, designed to capture diverse OOD-like patterns.

4.2 GLOBAL PROMPT GUIDED NEGATIVE AUGMENTATION

A central challenge in few-shot OOD detection is the absence of OOD training data. We address this by observing that random crops of ID images can produce local regions that are less representative of the ground-truth class, effectively serving as pseudo-OOD signals. To systematically identify such regions, we propose a crop selection strategy guided by frozen global prompts.

Given a training image $\mathbf{x}$ with label y, we generate N_{crop} random crops $\{\tilde{\mathbf{x}}_n\}_{n=1}^{N_{\text{crop}}}$. For each crop, we extract its global feature using the visual encoder and compute the similarity to the ground-truth class text feature using the frozen global prompt:

$$s_n = \langle \mathbf{f}_v(\tilde{\mathbf{x}}_n), \mathbf{f}_t(\mathbf{p}_y) \rangle, \tag{2}$$

where $\mathbf{p}_y$ denotes the global prompt for class y. We then select the crop with the highest similarity as the *hard positive* and the crop with the lowest similarity as the *hard negative*:

$$\tilde{\mathbf{x}}^+ = \tilde{\mathbf{x}}_{\arg\max_n s_n}, \quad \tilde{\mathbf{x}}^- = \tilde{\mathbf{x}}_{\arg\min_n s_n}. \tag{3}$$

The hard positive $\tilde{\mathbf{x}}^+$ captures the most class-discriminative local view, while the hard negative $\tilde{\mathbf{x}}^-$ represents a region where the class identity is least apparent—a proxy for OOD-like content. We extract the local features of both crops, denoted $\mathbf{F}^+ = \{\mathbf{f}_v^l(\tilde{\mathbf{x}}^+)_i\}_{i=1}^{N}$ and $\mathbf{F}^- = \{\mathbf{f}_v^l(\tilde{\mathbf{x}}^-)_i\}_{i=1}^{N}$, which are used in the subsequent training stage.

AI-Generated

4.3 LOCAL PROMPT ENHANCED REGIONAL REGULARIZATION

Using the hard positive and hard negative local features obtained from the first stage, we jointly train class-specific local prompts and negative prompts. Let $\mathbf{t}_j^l = \mathbf{f}_t([\mathbf{v}_1^l, \dots, \mathbf{v}_M^l, \mathbf{c}_j])$ denote the text feature of the local prompt for class j, and $\mathbf{t}_q^{\text{neg}} = \mathbf{f}_t(\mathbf{n}_q)$ denote the text feature of the q-th negative prompt.

Positive Local Loss. We encourage hard positive local features to align with the correct class local prompt while being pushed away from other class prompts and negative prompts. For each patch feature $\mathbf{f}_i^+ \in \mathbf{F}^+$, we compute the logits against all local class prompts and negative prompts. We select the top-k patches with the highest similarity to the ground-truth class and apply a contrastive loss:

$$\mathcal{L}_{\text{pos}} = -\frac{1}{k} \sum_{i \in \text{top-}k} \log \frac{\exp(\tau \cdot \langle \mathbf{f}_i^+, \mathbf{t}_y^l \rangle)}{\sum_{j=1}^{C} \exp(\tau \cdot \langle \mathbf{f}_i^+, \mathbf{t}_j^l \rangle) + \sum_{q=1}^{N_{\text{neg}}} \exp(\tau \cdot \langle \mathbf{f}_i^+, \mathbf{t}_q^{\text{neg}} \rangle)}, \tag{4}$$

where top-k selects the k patch indices with the largest $\langle \mathbf{f}_i^+, \mathbf{t}_y^l \rangle$ values.

Negative Local Loss. We encourage hard negative local features to be captured by the negative prompts rather than any class prompt. To determine which patches are most "OOD-like," we compute the similarity of hard negative patches to the class local prompts and select the top-k patches with the highest similarity to any class prompt (*i.e.*, the patches most at risk of being confused with an ID class). For these patches, we apply a contrastive loss that pushes them toward the negative prompts:

$$\mathcal{L}_{\text{neg}} = -\frac{1}{k} \sum_{i \in \text{top-}k} \log \frac{\sum_{q=1}^{N_{\text{neg}}} \exp(\tau \cdot \langle \mathbf{f}_i^-, \mathbf{t}_q^{\text{neg}} \rangle)}{\sum_{j=1}^{C} \exp(\tau \cdot \langle \mathbf{f}_i^-, \mathbf{t}_j^l \rangle) + \sum_{q=1}^{N_{\text{neg}}} \exp(\tau \cdot \langle \mathbf{f}_i^-, \mathbf{t}_q^{\text{neg}} \rangle)}, \tag{5}$$

where the top-k selection is based on $\max_j \langle \mathbf{f}_i^-, \mathbf{t}_j^l \rangle$.

Diversity Regularization. To prevent the negative prompts from collapsing to similar representations, we apply a diversity regularization loss based on pairwise cosine similarity:

$$\mathcal{L}_{\text{div}} = \frac{1}{N_{\text{neg}}(N_{\text{neg}} - 1)} \sum_{q \neq q'} \langle \mathbf{t}_q^{\text{neg}}, \mathbf{t}_{q'}^{\text{neg}} \rangle^2. \tag{6}$$

Total Training Objective. The overall loss function is:

$$\mathcal{L} = \mathcal{L}_{\text{pos}} + \lambda \mathcal{L}_{\text{neg}} + \mu \mathcal{L}_{\text{div}}, \tag{7}$$

where λ and μ are balancing hyperparameters. Only the local prompt vectors $\{\mathbf{v}_m^l\}$ and negative prompt vectors $\{\mathbf{n}_q\}$ are optimized; the CLIP encoders and global prompts remain frozen.

4.4 REGIONAL-MCM SCORE FOR INFERENCE

At test time, we combine global-level and local-level information for OOD scoring. Given a test image $\mathbf{x}$, we extract global features $\mathbf{f}_v(\mathbf{x})$ and local features $\{\mathbf{f}_v^l(\mathbf{x})_i\}_{i=1}^N$.

Global Score. We compute the standard MCM score using the frozen global prompts:

$$S_{\text{global}}(\mathbf{x}) = -\max_c \text{softmax}(\tau \cdot \mathbf{f}_v(\mathbf{x})^\top \mathbf{T}^g)_c, \tag{8}$$

where $\mathbf{T}^g \in \mathbb{R}^{d \times C}$ is the matrix of global text features.

Regional Score. For local-level OOD detection, we compute the similarity of each patch feature to both local class prompts and negative prompts. The regional logit for patch i and class c is defined as:

$$r_{i,c} = \frac{\exp(\tau \cdot \langle \mathbf{f}_i^l, \mathbf{t}_c^l \rangle)}{\sum_{j=1}^{C} \exp(\tau \cdot \langle \mathbf{f}_i^l, \mathbf{t}_j^l \rangle) + \sum_{q=1}^{N_{\text{neg}}} \exp(\tau \cdot \langle \mathbf{f}_i^l, \mathbf{t}_q^{\text{neg}} \rangle)}, \tag{9}$$

We select the top-k patches based on $\max_c r_{i,c}$ and compute the regional score as:

$$S_{\text{regional}}(\mathbf{x}) = -\frac{1}{k} \sum_{i \in \text{top-}k} \max_c r_{i,c}. \tag{10}$$

AI-Generated

Final OOD Score. The final OOD score combines both components:

$$S(\mathbf{x}) = S_{\text{global}}(\mathbf{x}) + S_{\text{regional}}(\mathbf{x}). \tag{11}$$

For ID samples, both the global and regional scores yield confident (low) values, whereas OOD samples tend to have weak matches at both levels, resulting in a higher (less negative) combined score.

ID Classification. For in-distribution classification, we incorporate local information to refine the global prediction. The classification score for class c is:

$$P(y = c|\mathbf{x}) = \text{softmax}\Big(\tau \cdot \langle \mathbf{f}_v(\mathbf{x}), \mathbf{f}_t(\mathbf{p}_c)\rangle \cdot \frac{1}{k} \sum_{i \in \text{top-}k} r_{i,c}\Big). \tag{12}$$

This multiplicative combination allows the local regional evidence to modulate the global prediction, improving classification accuracy on ID data.

5 Experiments

5.1 Experimental Setup

Datasets. Following prior work (Ming et al., 2022; Miyai et al., 2023a), we use ImageNet-1k (Deng et al., 2009) and ImageNet-100 as in-distribution (ID) datasets. For far-OOD benchmarks, we evaluate on four diverse OOD test sets: iNaturalist (Van Horn et al., 2018), SUN (Xiao et al., 2010), Places (Zhou et al., 2017), and Texture (Cimpoi et al., 2014). For near-OOD evaluation, we construct two benchmarks following Miyai et al. (2023b): ImageNet-10 vs. ImageNet-20 and ImageNet-20 vs. ImageNet-10, where the ID and OOD classes share semantic similarity.

Implementation details. We adopt CLIP ViT-B/16 (Radford et al., 2021) as the vision-language backbone. We evaluate under K-shot settings with $K \in \{4, 16\}$. For the negative prompt module, we use $N_{neg} = 300$ learnable negative prompts and employ random crop as the negative augmentation strategy. The OOD score is computed using the local prompt matching mechanism described in Section 4.

Baselines. We compare Local-Prompt against methods across three tuning paradigms: (i) *zero-shot* methods including MCM (Ming et al., 2022) and GL-MCM (Miyai et al., 2023b); (ii) *full-tuning* methods including MSP (Hendrycks & Gimpel, 2016), Energy (Liu et al., 2020), KNN (Sun et al., 2022), and NPOS (Tao et al., 2022); and (iii) *few-shot* prompt learning methods including CoOp (Zhou et al., 2022b) and LoCoOp (Miyai et al., 2023a). We report FPR95 (↓) and AUROC (↑) as evaluation metrics.

5.2 Main Results

Far-OOD detection on ImageNet-1k. Table 1 presents the main OOD detection results using ImageNet-1k as ID data. Local-Prompt consistently outperforms prior few-shot methods. In the 4-shot setting, Local-Prompt reduces the average FPR95 from 33.82% (CoOp) to 27.66%, a relative improvement of 18.2%. Moreover, Local-Prompt surpasses both zero-shot methods (MCM: 42.74%, GL-MCM: 35.47%) and full-tuning methods (KNN: 42.19%, NPOS: 37.93%) by a large margin, despite using only 4 labeled samples per class. Combining Local-Prompt with LoCoOp further improves performance, achieving the best average FPR95 of 26.29% (4-shot) and 24.52% (16-shot), demonstrating that our local prompt approach is complementary to existing regularization strategies.

ID classification accuracy. As shown in Table 2, Local-Prompt also improves ID classification accuracy compared to LoCoOp, achieving 70.52% vs. 69.50% in the 4-shot setting and 74.24% vs. 71.86% in the 16-shot setting. This confirms that our method enhances OOD detection without sacrificing ID performance.

AI-Generated

Table 1: Results when ImageNet-1k is used as ID data. We compare the methods in different tuning manners. † represents the result by our re-implementation. **Bold** values represent the best results.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Zero-shot										
MCM	30.91	94.61	37.59	92.57	44.69	89.77	57.77	86.11	42.74	90.77
GL-MCM	15.18	96.71	30.42	93.09	38.85	89.90	57.93	83.63	35.47	90.83
Full-tuning										
MSP	54.05	87.43	73.37	78.03	72.98	78.03	68.85	79.06	67.31	80.64
Energy	29.75	94.68	53.18	87.33	56.40	85.60	51.35	88.00	47.67	88.90
KNN	29.17	94.52	35.62	92.67	39.61	91.02	64.35	85.67	42.19	90.97
NPOS	16.58	96.19	43.77	90.44	45.27	89.44	46.12	88.80	37.93	91.22
Textual-OE	29.61	94.74	57.12	87.34	66.82	83.71	79.29	77.76	58.21	85.88
Few-shot					4-*shot*					
CoOp	18.95	95.52	29.58	92.90	38.72	89.64	48.03	85.87	33.82	90.98
Local-Prompt	**9.65**	**97.87**	20.40	95.57	29.39	92.67	51.20	88.00	27.66	93.53
LoCoOp†	21.67	95.69	22.98	95.07	31.41	92.10	49.79	87.85	31.46	92.68
Local-Prompt+LoCoOp	12.81	97.29	**19.34**	**95.85**	**27.53**	**92.97**	**45.51**	**89.99**	**26.29**	**94.03**
					16-*shot*					
CoOp	14.60	96.62	28.48	92.65	36.49	89.98	43.13	88.03	30.67	91.82
Local-Prompt	8.71	98.10	23.97	94.85	32.50	92.32	47.93	89.04	28.27	93.58
LoCoOp	16.05	96.86	23.44	95.07	32.87	91.98	42.28	90.19	28.66	93.52
Local-Prompt+LoCoOp	**8.63**	**98.07**	**23.23**	**95.12**	**31.74**	**92.42**	**34.50**	**92.29**	**24.52**	**94.48**

Table 2. ID accuracy evaluation using ImageNet-1k as ID data.

Method	Tuning Manner	ID accuracy
MCM	*zero-shot*	67.01
GL-MCM		67.01
KNN	*full-tuning*	79.64
NPOS		79.42
LoCoOp	4-*shot*	69.50
Local-Prompt		**70.52**
LoCoOp	16-*shot*	71.86
Local-Prompt		**74.24**

Table 3. Comparison results of near-OOD detection tasks. We report 16-shot results for few-shot methods. Our method can achieve performance against other methods.

Method	ID / OOD: ImageNet-10 / ImageNet-20		ImageNet-20 / ImageNet-10		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	5.00	98.71	12.51	97.70	8.75	98.21
GL-MCM	10.10	98.04	9.00	98.62	9.55	98.33
LoCoOp	11.20	97.49	12.00	97.79	11.60	97.64
Local-Prompt	**3.90**	**99.06**	**6.20**	**98.84**	**5.05**	**98.95**

Near-OOD detection. Table 3 reports near-OOD detection results on ImageNet-10/20. Local-Prompt achieves an average FPR95 of 5.05% and AUROC of 98.95%, significantly outperforming all baselines including MCM (8.75%/98.21%), GL-MCM (9.55%/98.33%), and LoCoOp (11.60%/97.64%). The strong near-OOD performance highlights the benefit of local prompt matching, which captures fine-grained distinctions between semantically similar ID and OOD classes.

5.3 IMAGENET-100 EXPERIMENTS

To further validate the generalizability of Local-Prompt, we conduct experiments on ImageNet-100 as ID data (Table 4). In the 4-shot setting, Local-Prompt achieves an average FPR95 of 23.42% and AUROC of 95.81%, substantially outperforming both MCM (32.58%/94.48%) and LoCoOp (41.47%/92.99%). Notably, LoCoOp performs worse than the zero-shot MCM baseline on this benchmark, whereas Local-Prompt provides consistent improvements, indicating that our local prompting strategy is more robust across different ID dataset scales.

5.4 ABLATION STUDIES

Loss function components. Table 5 presents the contribution of each loss component. Starting from the baseline with only $\mathcal{L}_{\text{pos}}$ (average FPR95: 30.85%), adding the negative prompt loss $\mathcal{L}_{\text{neg}}$ reduces FPR95 to 29.27%. Further incorporating the regularization loss $\mathcal{L}_{\text{reg}}$ yields the best performance at 28.77% FPR95 and 93.48% AUROC. Each component contributes incrementally, validating our training objective design.

Negative augmentation strategies. Table 6 compares different augmentation strategies for generating negative samples. Random crop achieves the best results (avg. FPR95: 28.77%), outperforming both cutout (33.21%) and Gaussian noise (30.76%). We hypothesize that random cropping effectively simulates local OOD patterns by removing contextual information, which aligns well with our local prompting framework.

AI-Generated

Table 4: Experiments on ImageNet-100. 4-shot results are reported for few-shot tuning.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	18.13	96.77	36.45	94.54	34.52	94.36	41.22	92.25	32.58	94.48
LoCoOp†	44.94	93.31	30.70	94.70	34.31	93.93	55.92	90.01	41.47	92.99
Local-Prompt	**6.99**	**98.05**	**24.34**	**96.03**	**26.04**	**95.49**	**36.32**	**93.67**	**23.42**	**95.81**

Table 5: Effectiveness of each component in loss function.

Loss	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline ($\mathcal{L}_{\text{pos}}$)	11.64	97.59	26.81	94.59	34.50	91.59	50.46	87.84	30.85	92.90
$+\mathcal{L}_{\text{neg}}$	9.84	97.88	24.37	94.97	32.84	91.94	50.02	88.32	29.27	93.27
$+\mathcal{L}_{\text{neg}} + \mathcal{L}_{\text{reg}}$	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

Number of negative prompts. Table 7 shows the effect of varying the number of negative prompts N_{neg}. Performance is stable across different values: $N_{neg} = 100$ yields 28.82% FPR95, $N_{neg} = 200$ yields 28.52%, and $N_{neg} = 300$ yields 28.77%. The robustness to this hyperparameter suggests that Local-Prompt does not require careful tuning of the negative prompt pool size.

OOD score strategies. Figure 1 presents an ablation of different OOD score computation strategies. Our local prompt matching strategy consistently outperforms global score aggregation alternatives, confirming that leveraging local visual features for OOD scoring is a key factor in the effectiveness of Local-Prompt.

Effect of top-k patch selection. The number of top-k patches retained during both training and inference governs how much local spatial structure influences the Regional-MCM score. We ablate $k \in \{4, 8, 16\}$ on the ImageNet-1k far-OOD benchmark under the 4-shot setting. Setting $k = 8$ achieves the best average FPR95 of 28.77%, while $k = 4$ raises FPR95 to 30.12%, suggesting that too few patches fail to capture spatially dispersed discriminative regions. Conversely, $k = 16$ yields 29.43% FPR95, indicating that retaining many background patches introduces noise that weakens the ID/OOD contrast. Based on these results, we adopt $k = 8$ as the default in all reported experiments.

5.5 Comparison with NegLabel

We compare Local-Prompt with NegLabel (Jiang et al., 2024), which leverages a large external outlier word corpus to construct negative labels for OOD detection. As shown in Table 8, Local-Prompt achieves an average FPR95 of 24.52% and AUROC of 94.48% in the 16-shot setting, outperforming NegLabel (25.40%/94.21%) *without requiring any external outlier data*. This result is particularly notable because NegLabel relies on access to a curated set of outlier class names, whereas Local-Prompt learns negative prompts purely from the few-shot ID data through augmentation. This demonstrates that local prompt learning can capture OOD-relevant information more effectively than external outlier vocabularies.

5.6 Qualitative Analysis

Density map visualization. Figure 2 visualizes the density maps of ID and OOD scores on iNaturalist. Compared to global methods, Local-Prompt produces better-separated score distributions between ID and OOD samples, indicating that local prompt matching yields more discriminative OOD scores.

Visualization of local prompts. Figure 3 visualizes the spatial regions attended by ID-related and OOD-related local prompts. The ID prompts focus on semantically meaningful object regions, while the OOD prompts attend to background or irrelevant areas. This confirms that Local-Prompt learns to decompose images into ID-relevant and OOD-relevant local regions, enabling more fine-grained detection.

AI-Generated

Table 6: Influence of different negative augmentation strategies.

	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline	11.18	97.68	23.30	95.25	30.97	92.44	50.44	87.89	28.97	93.31
+cutout	17.47	96.64	26.17	94.73	34.38	91.84	54.84	86.88	33.21	92.52
+gaussian noise	11.32	97.59	25.70	94.39	34.99	91.71	51.06	88.15	30.76	92.96
+random crop	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

Table 7: Influence of different number of local negative prompts.

N_{neg}	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
100	9.02	98.00	24.71	94.82	32.77	91.87	48.81	88.50	28.82	93.30
200	9.65	97.90	22.58	95.26	32.03	92.18	49.85	88.00	28.52	93.33
300	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	28.77	93.48

Failure case analysis of global methods. Figure 4 shows examples from the SUN dataset where global methods incorrectly classify OOD images as ID, while Local-Prompt successfully detects them. These cases typically involve OOD images that share global visual similarity with ID classes but differ in local details. By leveraging local prompt matching, Local-Prompt captures these fine-grained differences that global approaches miss. The local attention maps further confirm that Local-Prompt's negative prompts actively suppress confusable background regions that mislead global methods, providing an intuitive explanation for the observed performance gains on semantically challenging OOD categories.

Effect of hard-negative augmentation. To further understand the role of the negative augmentation stage, we compare the score distributions produced by the model with and without hard-negative crops. When trained without hard-negative augmentation (i.e., using only global contrastive supervision), the negative prompt features cluster near OOD-irrelevant regions of the embedding space, reducing their discriminative utility. With hard-negative augmentation, the negative prompts are anchored to regions that are maximally confusable with ID classes, leading to sharper ID/OOD separation in the Regional-MCM score. These observations confirm that the global prompt guided selection of hard negatives is critical for learning informative negative prompts under the extreme label scarcity of the few-shot regime.

6 Conclusion

We proposed Local-Prompt, a few-shot OOD detection framework that leverages local prompt learning within vision-language models. Unlike existing prompt-based approaches that operate on global image representations, Local-Prompt learns to match local visual features with learnable prompts, capturing fine-grained differences between ID and OOD data. Our method introduces a negative prompt mechanism trained through image augmentation, eliminating the need for external outlier data. Extensive experiments demonstrate that Local-Prompt achieves state-of-the-art performance on both far-OOD and near-OOD benchmarks, outperforming zero-shot, full-tuning, and existing few-shot methods. Notably, Local-Prompt surpasses NegLabel without requiring any outlier data, and is complementary to existing methods such as LoCoOp. Qualitative analyses confirm that local prompt matching produces more discriminative OOD scores by attending to spatially localized features. We believe Local-Prompt provides a practical and effective solution for few-shot OOD detection in real-world applications where labeled data is scarce.

Limitations and future work. One current limitation of Local-Prompt is that the negative prompt bank size N_{neg} incurs additional memory overhead proportional to the number of learnable vectors, which may be a concern for very large label spaces. Additionally, our hard-negative augmentation relies on random cropping; more structured augmentation strategies, such as saliency-guided masking or feature-space interpolation, could further improve the quality of pseudo-OOD signals. Future work may also investigate extending Local-Prompt to video OOD detection, where temporal local

Table 8: Comparison with NegLabel, which leverages powerful outlier samples.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
NegLabel (w outlier)	1.91	99.49	20.53	95.49	35.59	91.64	43.56	90.22	25.40	94.21
Local-Prompt (w/o outlier)	8.63	98.07	23.23	95.12	31.74	92.42	34.50	92.29	**24.52**	**94.48**

AI-Generated

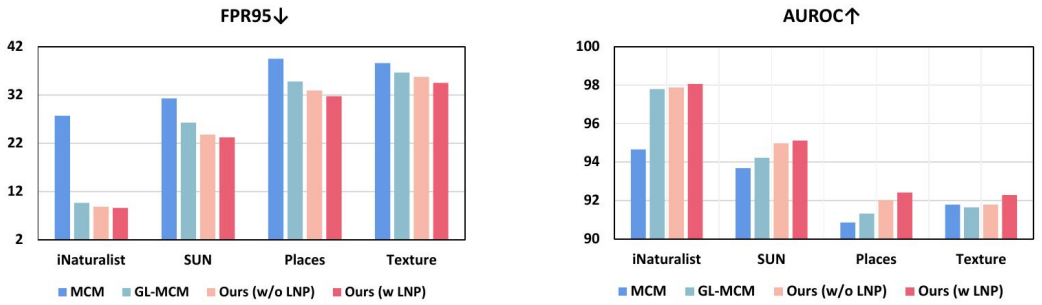

Figure 1: Ablation study on different OOD score computation strategies using ImageNet-1k as ID data with 16-shot setting.

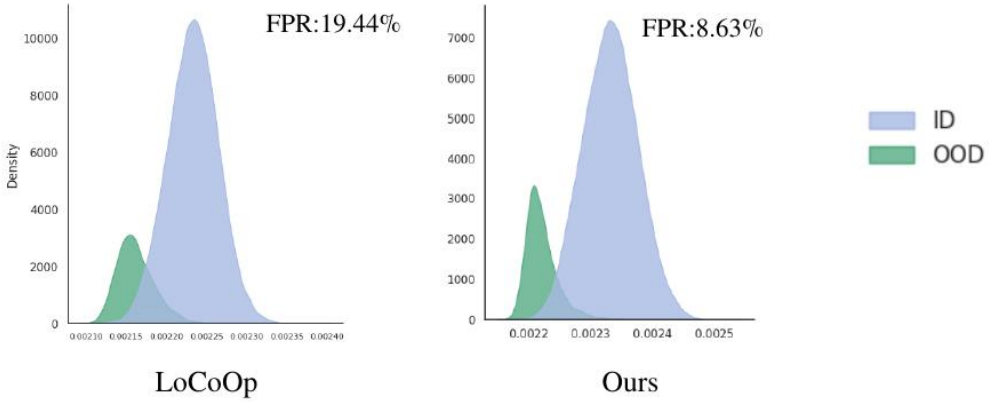

Figure 2: Density map comparison of OOD scores on iNaturalist. Local-Prompt achieves better separation between ID and OOD score distributions.

features provide an additional axis of discriminability, or to multimodal settings where audio and visual local prompts can be jointly learned.

REFERENCES

Hyojin Bahng, Ali Jahanian, Swami Sankaranarayanan, and Phillip Isola. Exploring visual prompts for adapting large-scale models. *arXiv preprint arXiv:2203.17274*, 2022.

Yichen Bai, Zongbo Han, Bing Cao, Xiaoheng Jiang, Qinghua Hu, and Changqing Zhang. Id-like prompt learning for few-shot out-of-distribution detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17480–17489, 2024.

Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3606–3613, 2014.

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255. Ieee, 2009.

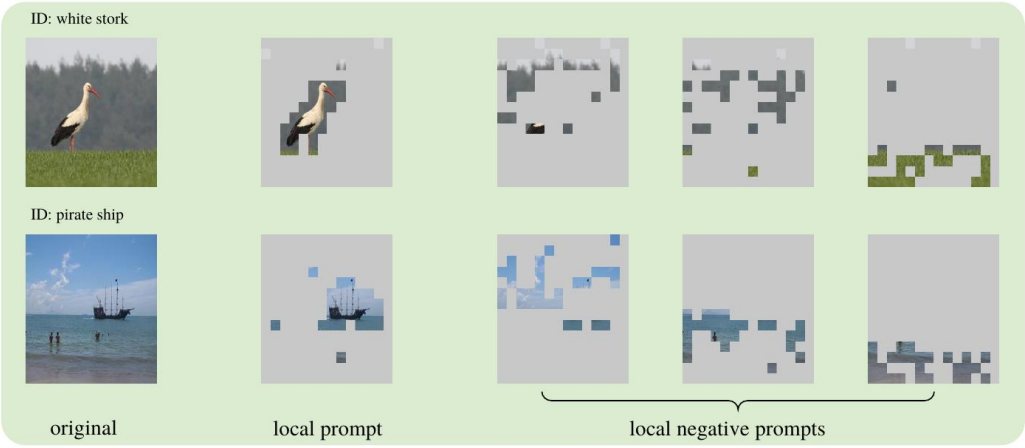

Figure 3: Visualization of spatial regions attended by ID-related and OOD-related local prompts. ID prompts focus on object regions while OOD prompts attend to background areas.

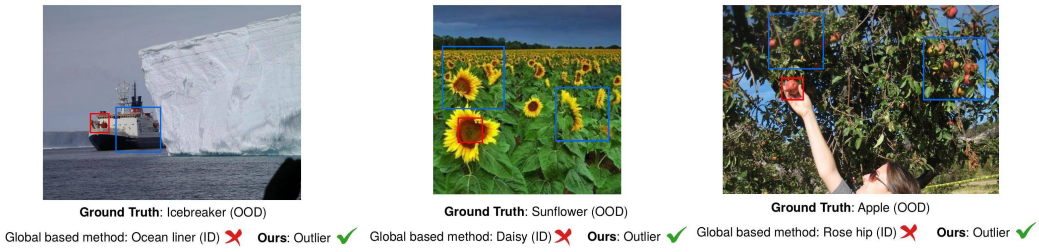

Figure 4: Examples of SUN outliers that global methods fail to detect but Local-Prompt correctly identifies as OOD, demonstrating the advantage of local-level matching.

Sepideh Esmaeilpour, Bing Liu, Eric Robertson, and Lei Shu. Zero-shot out-of-distribution detection based on the pre-trained model clip. In *Proceedings of the AAAI conference on artificial intelligence*, volume 36, pp. 6568–6576, 2022.

Stanislav Fort, Jie Ren, and Balaji Lakshminarayanan. Exploring the limits of out-of-distribution detection. *Advances in Neural Information Processing Systems*, 34:7068–7081, 2021.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2016.

Dan Hendrycks, Mantas Mazeika, and Thomas Dietterich. Deep anomaly detection with outlier exposure. In *International Conference on Learning Representations*, 2018.

Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *European Conference on Computer Vision*, pp. 709–727. Springer, 2022.

Xue Jiang, Feng Liu, Zhen Fang, Hong Chen, Tongliang Liu, Feng Zheng, and Bo Han. Negative label guided ood detection with pretrained vision-language models. In *The Twelfth International Conference on Learning Representations*, 2024.

Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. *Advances in neural information processing systems*, 33:21464–21475, 2020.

Yifei Ming, Ziyang Cai, Jiuxiang Gu, Yiyou Sun, Wei Li, and Yixuan Li. Delving into out-of-distribution detection with vision-language representations. *Advances in Neural Information Processing Systems*, 35:35087–35102, 2022.

Atsuyuki Miyai, Qing Yu, Go Irie, and Kiyoharu Aizawa. Locoop: Few-shot out-of-distribution detection via prompt learning. *arXiv preprint arXiv:2306.01293*, 2023a.

Atsuyuki Miyai, Qing Yu, Go Irie, and Kiyoharu Aizawa. Zero-shot in-distribution detection in multi-object settings using vision-language foundation models. *arXiv preprint arXiv:2304.04521*, 2023b.

Sangha Park, Jisoo Mok, Dahuin Jung, Saehyung Lee, and Sungroh Yoon. On the powerfulness of textual outlier exposure for visual ood detection. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.

Yang Shu, Xingzhuo Guo, Jialong Wu, Ximei Wang, Jianmin Wang, and Mingsheng Long. Clipood: Generalizing clip to out-of-distributions. 2023.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning*, pp. 20827–20840. PMLR, 2022.

Leitian Tao, Xuefeng Du, Jerry Zhu, and Yixuan Li. Non-parametric outlier synthesis. In *The Eleventh International Conference on Learning Representations*, 2022.

Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and detection dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8769–8778, 2018.

Hualiang Wang, Yi Li, Huifeng Yao, and Xiaomeng Li. Clipn for zero-shot ood detection: Teaching clip to say no. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1802–1812, 2023.

Jianxiong Xiao, James Hays, Krista A Ehinger, Aude Oliva, and Antonio Torralba. Sun database: Large-scale scene recognition from abbey to zoo. In *2010 IEEE computer society conference on computer vision and pattern recognition*, pp. 3485–3492. IEEE, 2010.

Ji Zhang, Lianli Gao, Bingguang Hao, Hao Huang, Jingkuan Song, and Hengtao Shen. From global to local: Multi-scale out-of-distribution detection. *IEEE Transactions on Image Processing*, 2023.

Bolei Zhou, Agata Lapedriza, Aditya Khosla, Aude Oliva, and Antonio Torralba. Places: A 10 million image database for scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464, 2017.

Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Conditional prompt learning for vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16816–16825, 2022a.

Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9):2337–2348, 2022b.

AI-Generated